Dimensionality Reduction

"*Art is the elimination of the unnecessary*"

By Pablo Picasso

Dimensionality Reduction

- Process of selecting a subset of features
- Pre-processing of data in machine learning
- Can be used in both Supervised and Unsupervised learning problems

Dimensionality Reduction

- Goal

Dimensional reduction method is to reduce the dimensions of the original data for different purposes such as visualization, decrease CPU time, ..etc..

- Application

-Machine learning

-Data mining

-Bioinformatics

-Biometric

-Information retrieval etc.…….

Dimensionality Reduction Technique

- Percent missing values
- Amount of variation
- Principal Component Analysis (PCA)
- Linear Discriminant Analysis (LDA)
- Generalized Discriminant Analysis (GDA)
    - We will focus on two methods that is PCA (Unsupervised) and LDA (Supervised)

Introduction To Principal Component Analysis (PCA)

- Principal Components (PCs)
  - Covariance Matrix Method
  - Singular Value Decomposition (SVD) Method
  - PCA Space (Lower Dimensional Space)
  - Data Reconstruction
  - PCA Algorithms
- Numerical example
  - First Example: 2D-Class Example
  - Multi-Class Example

Principal Components (PCs)

- The PCA space consists of *k* principal components.
- The principal components are orthonormal $\left[{v_i}^T v_j \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases} \right]$, uncorrelated $\left[Cov\left(v_i, v_j\right) = 0, where\ i \neq j \right]$, and it represents the direction of the maximum variance.
- The first principal component ($PC_1\ or\ v_1$) $\in R^{M \times 1}$) of the PCA space represents the direction of the maximum variance of the data, the second principal component has the second largest variance, and so on.

Principal Components (PCs)

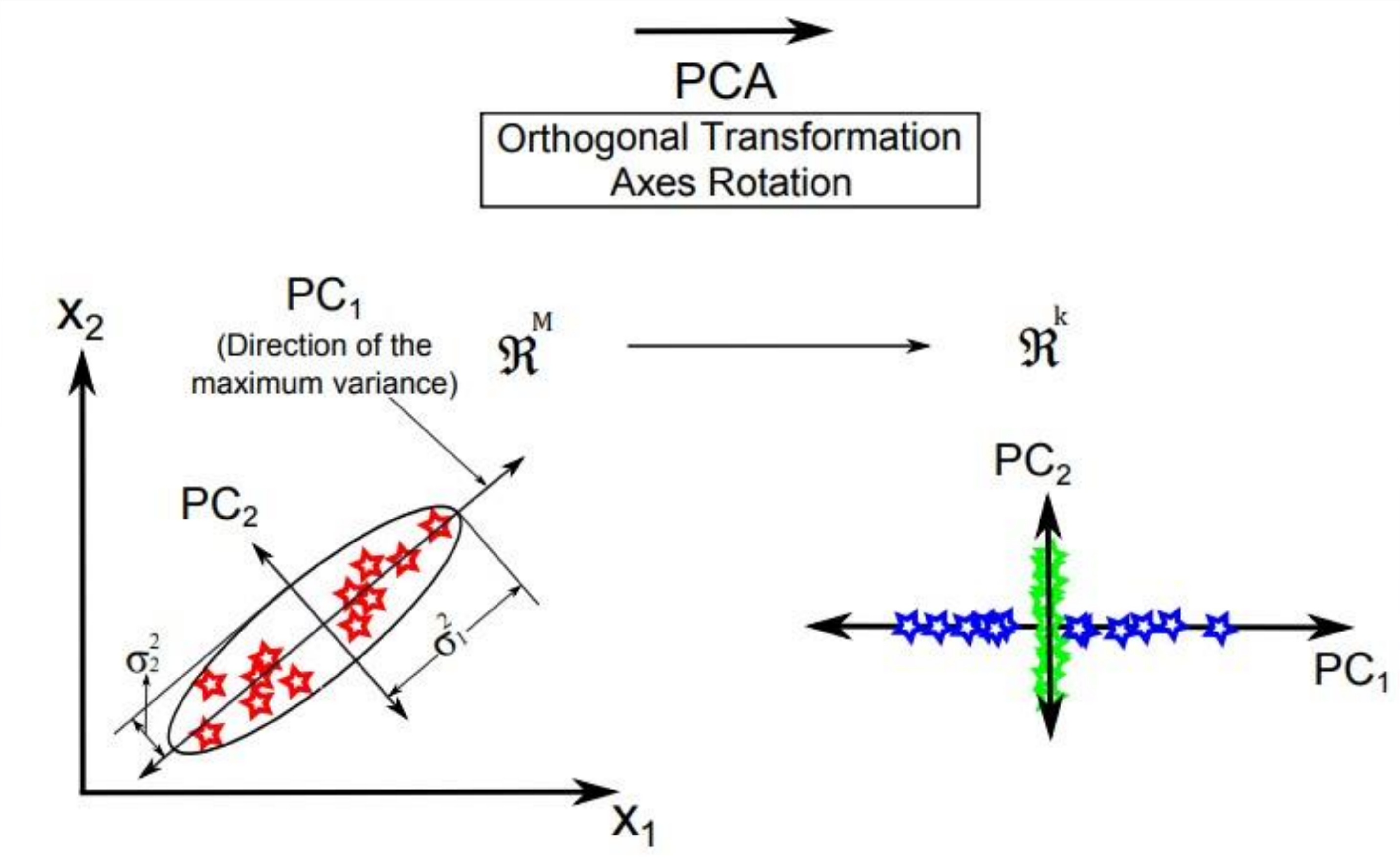


Fig: 1 Example of the two-dimensional data (x_1, x_2). The original data are on the left with the original coordinate, i.e. x_1 and x_2, the variance of each variable is graphically represented and the direction of the maximum variance, i.e. the principal component PC_1, is shown; on the right the original data are projected on the first (blue stars) and second (green stars) principal components [1].

Principal Components (PCs)

- There are two methods to calculate the principal components.

  - Covariance matrix method
  - Singular value decomposition (SVD) method.

Covariance Matrix Method

- In this method, there are two main steps to calculate the PCs of the PCA space.
  - The covariance matrix of the data matrix $(X)$is calculated.
  - The eigenvalues and eigenvectors of the covariance matrix are calculated.

Algorithm

Step 1: Calculate the mean of each variable in the data matrix, the mean-centering data are calculated by subtracting the mean from each sample.

Step 2: The covariance matrix is then calculated by , $\sum = DD^T$

Matrix formed by Covariance is symmetric matrix

$$\Sigma = \begin{pmatrix} Var(x_1, x_2) & Cov(x_1, x_2) \ldots \ldots .. & Cov(x_1, x_M) \\ Cov(x_2, x_1) & Var(x_2, x_2) \ldots \ldots .. & Cov(x_2, x_M) \\ . & . & . \\ . & . & . \\ . & . & . \\ Cov(x_M, x_1) & Cov(x_M, x_2) \ldots \ldots .. & Var(x_M, x_M) \end{pmatrix}$$

Algorithm

Step 3: Computing eigenvector v and eigenvalues λ of obtained Covariance matrix by equation given below.

$$v\sum = \lambda v$$

Step 4: Sort eigenvectors according to their corresponding eigenvalues.

Step 5: The eigenvector with the highest eigenvalue (W) represents the first principal component and it has the maximum variance.

$$W = \{v_1, v_2, v_3 \ldots\ldots, v_k\}$$

Step 5: All samples are projected on the lower dimensional space of PCA (W) as follows

$$Y = W^T D$$

Algorithm

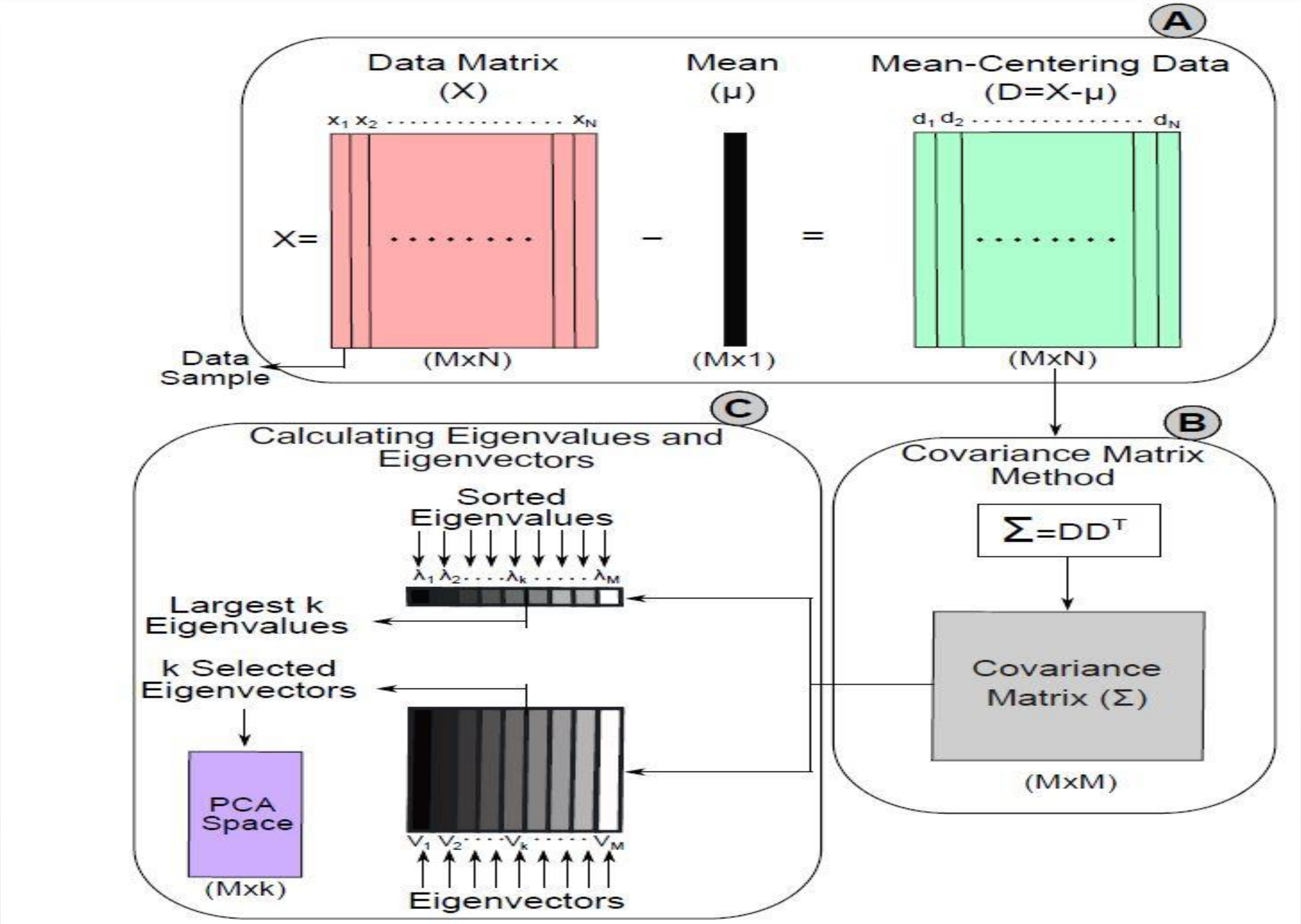


Fig: 2 Visualized steps to calculate the PCA space using the covariance matrix method [1].

Example:

Lets have a small data matrix

X=

X_1	X_2
1	3
1	2
2	3
0	3
5	4
4	5
5	5
3	4

Solution of the Example:

- Step 1: calculate mean　　Step 2: subtracting from the mean

X_1	X_2
1	3
1	2
2	3
0	3
5	4
4	5
5	5
3	4
$\mu_1 = 2.63$	$\mu_2 = 3.63$

$X_1 - \mu_1$	$X_2 - \mu_2$
1-2.63	3-3.63
1-2.63	2-3.63
2-2.63	3-3.63
0-2.63	3-3.63
5-2.63	4-3.63
4-2.63	5-3.63
5-2.63	5-3.63
3-2.63	4-3.63

D=

$X_1 - \mu_1$	$X_2 - \mu_2$
-1.63	-0.63
-1.63	-1.63
-0.63	-0.63
-2.63	-0.63
2.38	0.38
1.38	1.38
2.38	1.38
0.38	0.38

Solution

Step. 3 Compute Covariance matrix	Step. 4 Compute eigenvalues	Step. 5 Compute eigenvector
Using $\sum = \frac{1}{N-1}(DD^T)$	Using D-λ[I]=0 where D is covariance matrix, I is unit matrix=$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ Solving it.	Using D-λ[I][v]=0 for Each λ value solving it we get

Covariance matrix:

3.70	1.70
1.70	1.13

λ=

0.28	0
0	4.54

V=

0.45	-0.90
-0.90	-0.45

From the results, we note that the second eigenvalue (λ_2) was more than the first one (λ_1).

Singular Value Decomposition (SVD) Method

- Singular value decomposition is one of the most important linear algebra principles. The aim of the SVD method is to diagonalize the data matrix ($X \epsilon R^{p \times q}$) into three matrices.
- left singular vectors $L(p \times p)$, diagonal matrix $S(p \times q)$ represents the singular values that are sorted from high-to-low, i.e. the highest singular value in the upper-left index of S, thus $s_1 \geq s_2 \dots \dots \geq s_q \geq 0$, and right singular vectors R$(p \times q)$.

$$X = LSR^T = [l_1 \dots \dots \dots . l_m] \begin{bmatrix} s_1 & 0 & 0 \\ 0 & s_2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 \cdots & s_N \\ \vdots & \ddots & \vdots \\ 0 & 0 \cdots & 0 \end{bmatrix} \begin{bmatrix} -r_1^T - \\ -r_2^T - \\ . \\ . \\ . \\ -r_N^T - \end{bmatrix}$$

Singular Value Decomposition (SVD) Method

- To calculate SVD, R^T and S are first calculated by diagonalizing $X^T X$ as follows,

$$X^T X = (LSR^T)^T \ (LSR^T)$$

$$= L^T S^T RLSR^T$$

$$= \mathrm{R}\ S^2 R^T \qquad ,(L^T L = \mathrm{I})$$

- The left singular vectors (L) is then calculated as follows, $L \ = \ XRS^{-1}$, where X_{r_i} is in the direction of $s_i l_i$.
- (R) represent the eigenvectors of $X^T X$ or the principal components of the PCA space, and ${s_i}^2, \forall_i = 1,2, \ldots \ldots \ldots q$ represent their corresponding eigenvalues.

Singular Value Decomposition (SVD) Method

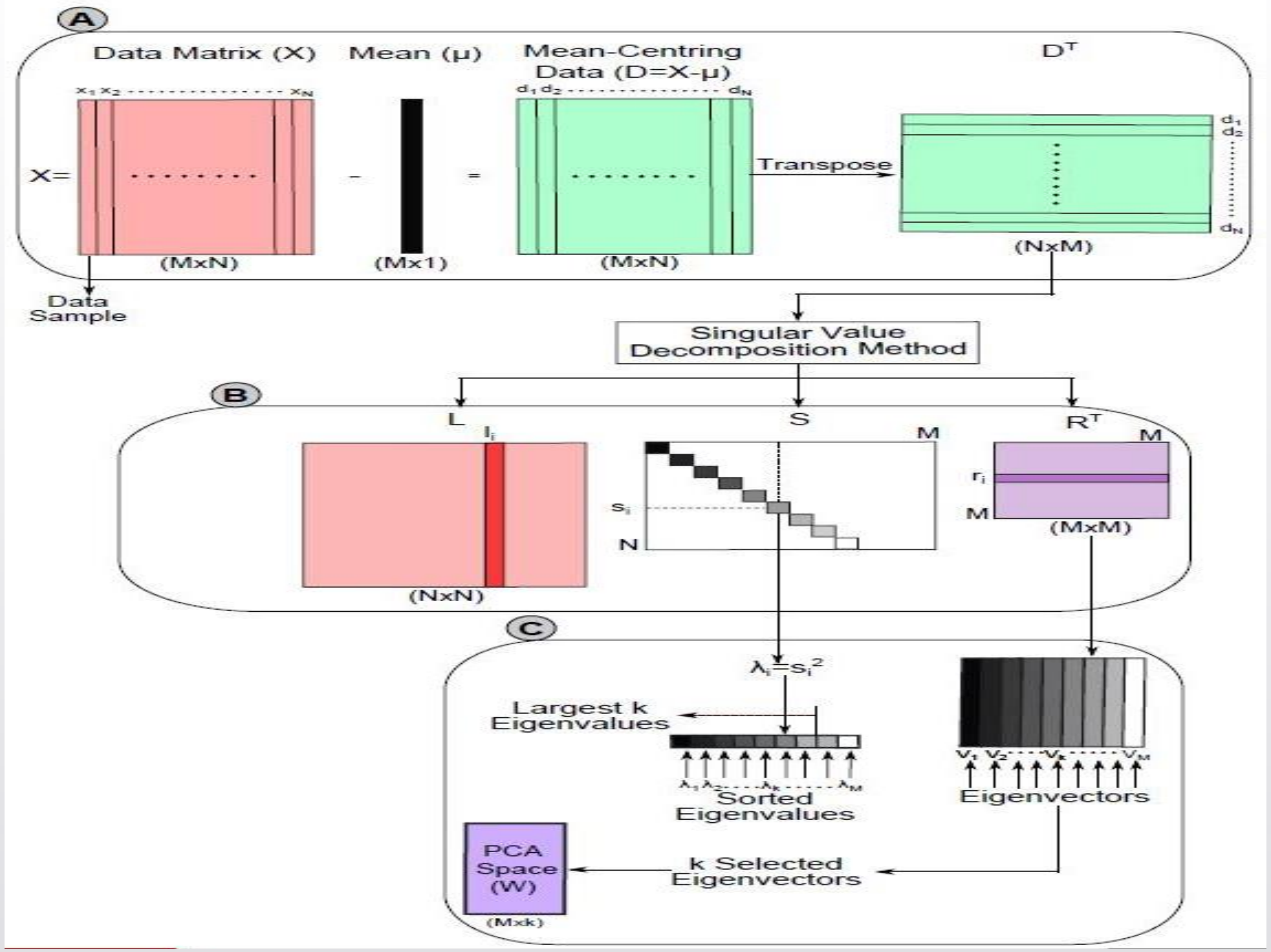


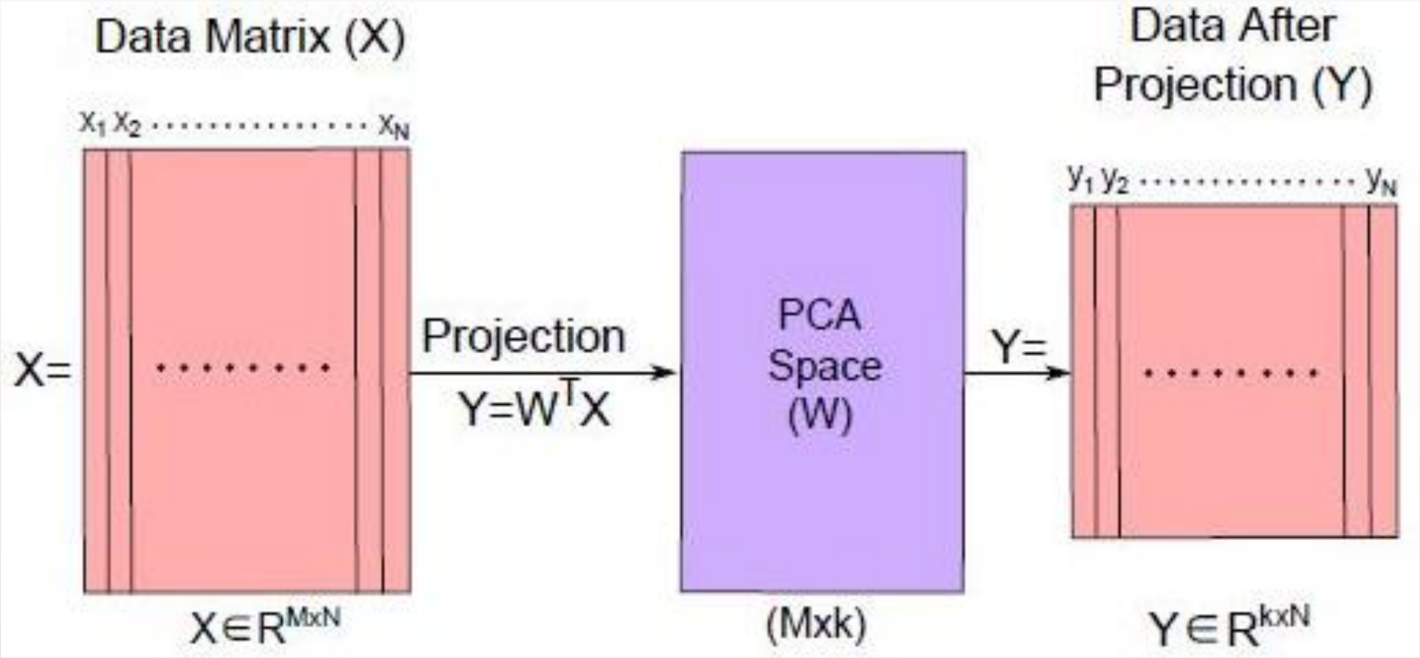
Fig: 3 Visualized steps to calculate the PCA space using the SVD method [1].

PCA Space (Lower Dimensional Space)

- To construct the lower dimensional space of PCA (W), a linear combination of k selected PCs that have the most k eigenvalues are used to preserve the maximum amount of variance, i.e. preserve the original data, while the other eigenvectors or PCs are neglected.
- The lower dimensional space is denoted by $W = \{v_{1,} v_{2,} \ldots \ldots v_{k,}\}$. The dimension of the original data is reduced by projecting it after subtracting the mean onto the PCA space as follows:

$$Y = W^T D = \sum_{i=1}^{N} W^T (x_i - \mu)$$

- Here Y is the original data after projecting it onto the PCA space; thus; $(M - k)$ features or variables are lost from the original data.


Data Matrix (X)
x1 x2 .... xN
X=
X∈R^MxN
Projection
Y=W^TX
PCA
Space
(W)
(Mxk)
Y=
Data After
Projection (Y)
y1 y2 .... yN
Y∈R^kxN

Data Reconstruction

- The original data can be reconstructed again as follows:

$$\hat{X} = WY + \mu = \sum_{i=1}^{N} W y_i + \mu$$

where $\hat{X}$ is reconstructed data

- The deviation between the original data and the reconstructed data are called the reconstruction error or residuals as follows:

$$error = X - \hat{X} = \sum_{i=1}^{N} (x_i - \hat{x}_i)^2$$

Linear Discriminant Analysis (LDA)

- Introduction to Linear Discriminant Analysis (LDA).
- Definition of LDA.
    - Calculating the Between-Class Variance (SB).
    - Calculating the Within-Class Variance (SW).
    - Constructing the Lower Dimensional Space.
    - Projection onto the LDA space.
    - Example of Two LDA Subspaces.
    - Computational Complexity of LDA.
    - Class-Dependent vs. Class-Independent Methods.
- Numerical Example.
- Conclusions.

The Linear Discriminant Analysis (LDA)

- The goal of the LDA technique is to project the original data matrix onto a lower dimensional space.
- To achieve this goal, three steps needed to be performed.

Step 1: calculate the separability between different classes which is called the **between-class variance or between-class matrix.**

Step 2: calculate the distance between the mean and the samples of each class, which is called the **within-class variance or within-class matrix.**

Step 3: maximizes the between-class variance and minimizes the within-class variance.

Calculating the Between-Class variance (SB)

- Computed By $(m_i - m)^2$

$$=(W^T\mu_i - W^T\mu)^2$$

$$=W^T(\mu_i - \mu)\ (\mu_i - \mu)^T W$$

$$= W^T S_{B_i} W$$

- Where, m_i- the projection of the mean of the i^{th}class

  m- the projection of the total mean of all classes

  W-transformation matrix of LDA

  μ_i- the mean of the i^{th} class

  $\mu -$ total mean of all classes

  $S_{B_i} = (\mu_i - \mu)\ (\mu_i - \mu)^T$- separation distance between the mean of the i^{th} class (μ_i) and the total mean (μ),

Calculating the Within-Class variance (SW)

- minimize the difference between the projected mean (m_i) and the projected samples of each class ($W^T x_i$), or simply minimizes the within-class variance.

$$= \sum_{x_i \epsilon w_j} W^T (x_i - \mu_j)\,(x_i - \mu_j)^T W$$

$$= \sum_{x_i \epsilon w_j} W^T S_{W_j} W$$

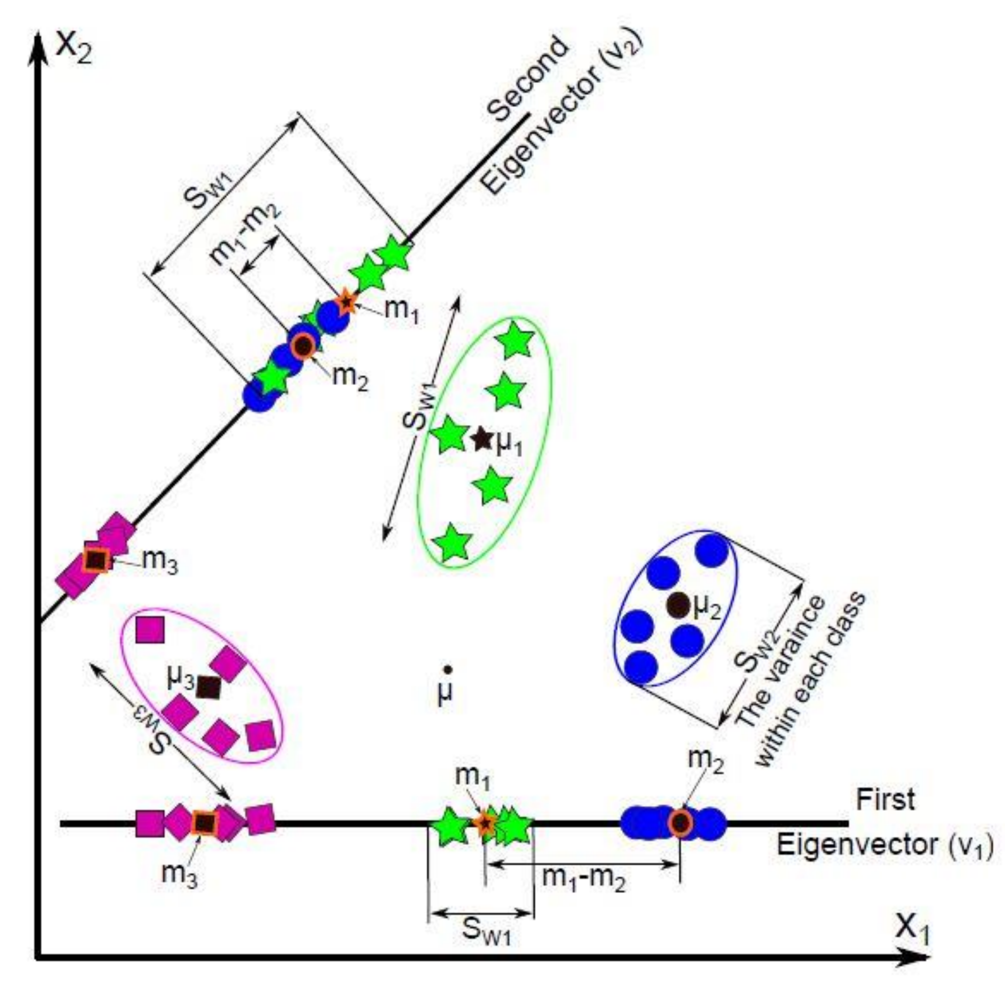


Fig. 4 A visualized comparison between the t w o lower -dimensional sub-spaces which are calculated using three different classes [2].

Constructing the lower dimensional space

- After calculating the between-class variance (S_B) and within-class variance (S_W), the transformation matrix (W) of the LDA technique can be calculated as

$$\arg\max_W \frac{W^T S_B W}{W^T S_W W} \quad \rightarrow S_W W = \lambda S_B W$$

Where λ- eigen Values of the transformation matrix (W)

- The solution of this problem can be obtained by calculating the eigenvalues (λ) and eigenvectors ($V = \{V_1, V_2 \ldots \ldots . V_M\}$) of $W = S_W^{-1} S_B$, if S_W is non-singular.

- the dimension o f the original data matrix ($X \epsilon R^{N \times M}$) is reduced by projecting it onto the lower dimensional space of LDA ($V_k \epsilon R^{M \times k}$) as The dimension of the data after projection is k ; hence, $M - k$ features are ignored or deleted from each sample.

Numerical example

Given two different classes, $W_1(5\times2)$ and $W_2(6\times2)$ have $(n_1 = 5)$ and $(n_2 = 6)$ samples, respectively. Each sample in both classes is represented by two features (i.e. M = 2) as follows:

- $w_1 = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 3 \\ 4 & 5 \\ 5 & 5 \end{bmatrix}_{5\times2}$ $w_2 = \begin{bmatrix} 4 & 2 \\ 5 & 0 \\ 5 & 2 \\ 3 & 2 \\ 5 & 3 \\ 6 & 3 \end{bmatrix}_{6\times2}$

Solution

- First compute means of W_1 and W_2

$$\mu_1 = [3 \quad 3.60], \mu_2 = [4.67 \quad 2] \text{ and total mean value } \mu = \left[\frac{5}{11}\mu_1 \quad \frac{6}{11}\mu_2\right]$$

$$= [3.91 \quad 2.727]$$

- To Calculate $S_{B_1} = n_1(\mu_1 - \mu)^{sT}(\mu_1 - \mu) = 5[-0.91 \quad 0.87]^T[-0.91 \quad 0.87]$

$$= \begin{bmatrix} 4.13 & -3.97 \\ -3.97 & 3.81 \end{bmatrix}$$

Similarity for $S_{B_2} = \begin{bmatrix} 3.44 & -3.31 \\ -3.31 & 3.17 \end{bmatrix}$

- The total between-class variance is calculated a follows: $S_B = S_{B_1} + S_{B_2}$

$$S_B = \begin{bmatrix} 7.58 & -7.27 \\ -7.27 & 6.98 \end{bmatrix}$$

- To calculate S_W, first calculate mean-centering data.
  $d_1 = W_1 - \mu_1$ and $d_2 = W_2 - \mu_2$

$$d_1 = \begin{bmatrix} -2 & -1.60 \\ -1 & -0.60 \\ 0 & -0.60 \\ 1 & 1.40 \\ 2 & 1.40 \end{bmatrix} \text{ and } d_2 = \begin{bmatrix} -0.67 & 0 \\ 0.33 & -2.00 \\ 0.33 & 0.00 \\ -1.67 & 0.00 \\ 0.33 & 1.00 \\ 1.33 & 1.00 \end{bmatrix}$$

After centering the data, in class-independent method, the within-class variance for each class is calculated as follows

$$S_{W_i} = d_i^T \times d_i$$

$$S_{W_1} = \begin{bmatrix} 10 & 8 \\ 8 & 7.20 \end{bmatrix} \text{ and } S_{W_2} = \begin{bmatrix} 5.33 & 1 \\ 1 & 6 \end{bmatrix}, \ S_w = S_{W_1} + S_{W_2} = \begin{bmatrix} 15.33 & 9 \\ 9 & 13.20 \end{bmatrix}$$

The transformation matrix (W) can be obtained as follows, $W = S_W^{-1} S_B$, and the values of $(S_W^{-1})$and (W) are as follows:

$$S_W^{-1} = \begin{bmatrix} 0.11 & -0.07 \\ -0.07 & 0.13 \end{bmatrix} \text{and } W = \begin{bmatrix} 1.37 & -1.32 \\ -1.49 & 1.43 \end{bmatrix}$$

The eigenvalues and eigenvectors of W are then calculated as follows:

$$\lambda = \begin{bmatrix} 0.00 & 0.00 \\ 0.00 & 2.81 \end{bmatrix} \text{ and V} = \begin{bmatrix} -0.69 & 0.68 \\ -0.72 & -0.74 \end{bmatrix}$$

The second eigenvector (V_2) has corresponding eigenvalue more than the first one (V_1), which reflects that, the second eigenvector is more robust than the first one; hence, it is selected to construct the lower dimensional space.

Advantages of Dimensionality Reduction

- helps in data compression, and hence reduced storage space.
- reduces computation time.
- remove redundant features
- It takes care of multicollinearity that improves the model performance.

Disadvantages of Dimensionality Reduction

- lead to some amount of data loss.
- PCA tends to find linear correlations between variables, which is sometimes undesirable.

References

[1] Tharwat, A. (2016). Principal component analysis-a tutorial. IJAPR, 3(3), 197-240.

[2] Tharwat, A., Gaber, T., Ibrahim, A., & Hassanien, A. E. (2017). Linear discriminant analysis: A detailed tutorial. AI communications, 30(2), 169-190.

Thanking You